Operation Management

BITS Pilani
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<Operation Management>
Lecture No. 8

Human Resource Strategy

The study material is prepared by Prof. Abhishek Maiti for the students of Operation Management

- Labor planning.
- Job design.
- Labour standards.

Human Resource Strategy

- **Labour Planning** — Employment Stability, Work Schedules, Work Rules
- **Job Design** — Job Specialization, Job Expansion, Psychological components, Self Directed Teams, Motivation and Incentive System
- **Labour Standards** — Historical experiences, Time Studies, Predetermined standards, Work Sampling

Labour Planning

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- Employment stability.
- Work schedules.
- Work rules.

Employment Stability

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- No. of employees maintained by an organization at any given time.

There are two very basic policies to do so :

- Follow demand exactly.
- Hold employment constant.

Work Schedules

The study material is prepared by Prof. Abhishek Maiti for the students of Operation Management

- Fixed job timing.
- Flextime.
- Flexible work week or Compressed work week.
- Shift jobs.
- Part time.

Job Classification & Work Rules

The study material is prepared by Prof. Abhishek Maiti for the students of Operation Management

- Formalization.
- Normalization.

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Any Question

Human Resource Strategy

Labour Planning

Employment Stability, Work Schedules, Work Rules

Job Design

Job Specialization, Job Expansion, Psychological components, Self Directed Teams, Motivation and Incentive System

Labour Standards

Historical experiences, Time Studies, Predetermined standards, Work Sampling

Job Design

The study material is prepared by Prof. Abhishek Maiti for the students of Operation Management

- Referred as Work Design in many books.
- It refers how the work is structured, Organized, Experienced and Enacted.
- There are five components of job design: (1) job specialization, (2) job expansion, (3) psychological components, (4) self-directed teams, and (5) motivation and incentive systems.

Labour Specialization

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- Referred as Job Specialization in many books.
- The division of labour into unique (“special”) tasks.

Source: Operation Management by William J Stevenson

Job Expansion

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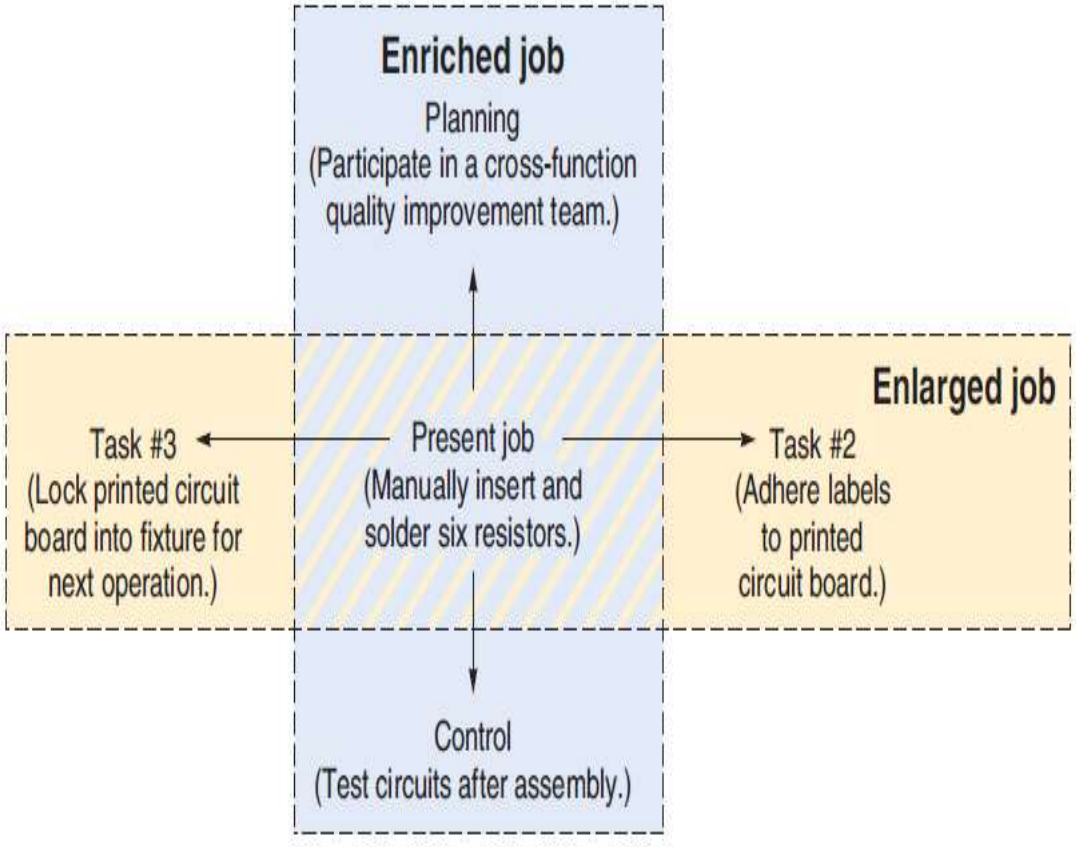

- Job Enlargement
- Job Rotation
- Job Enrichment
- Employee Empowerment

Source: Heizer, Render, Muckson

Psychological Components

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- Hawthorne studies.
- Jobs should be designed in such a way that it includes Skill Variety, Job identity, Job significance, autonomy and feedback.

Job Characteristics Model

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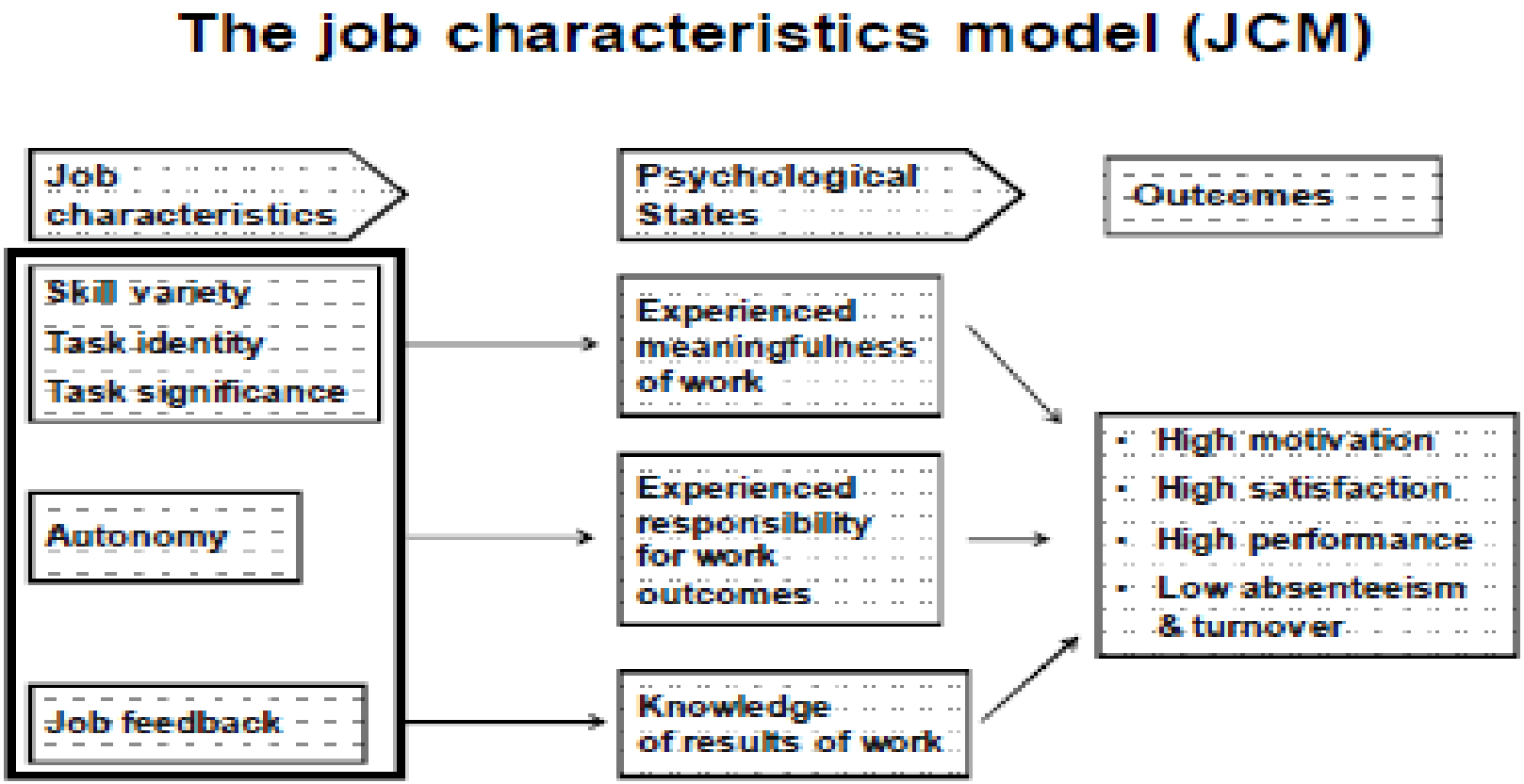

Self Directed Teams

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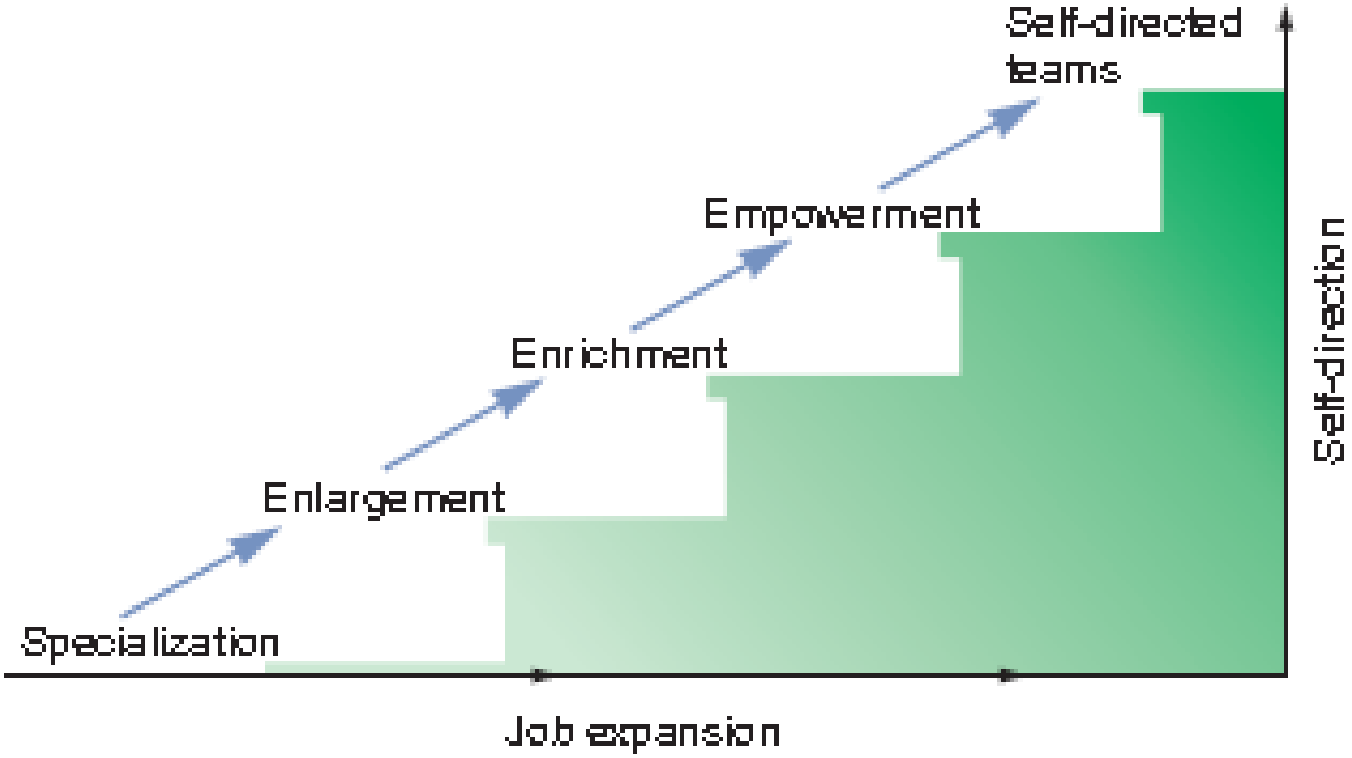

Source: Heizer, Render, Muckson

Motivation and Incentive Systems

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- Bonuses – cash or stock options.
- Profit-sharing – profits for distribution to employees.
- Gain sharing – rewards for improvements.
- Incentive systems – typically based on production rates.
- Knowledge-based systems – reward for knowledge or skills.

Hard Rock Cafe has successfully reduced its turnover by giving every employee—from the CEO to the busboys—a $10,000 gold Rolex watch on their 10th anniversary with the firm.

Ergonomics and the Work Environment

- Ergonomics is the study of the human interface with the environment and machines
  - Often called *human factors*
- Operator input to machines needs to be carefully evaluated

Ergonomics and Work Methods

- Feedback to operators
- The work environment
  - *Illumination*
  - *Noise*
  - *Temperature*
  - *Humidity*

Method Analysis

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The techniques used in the Method Analysis involves:

- Movement of individuals or material : Flow Diagram and Process chart.
- Activity of human and machine and crew activity : Activity chart.
- Body movement : Operation chart.

Flow Diagram

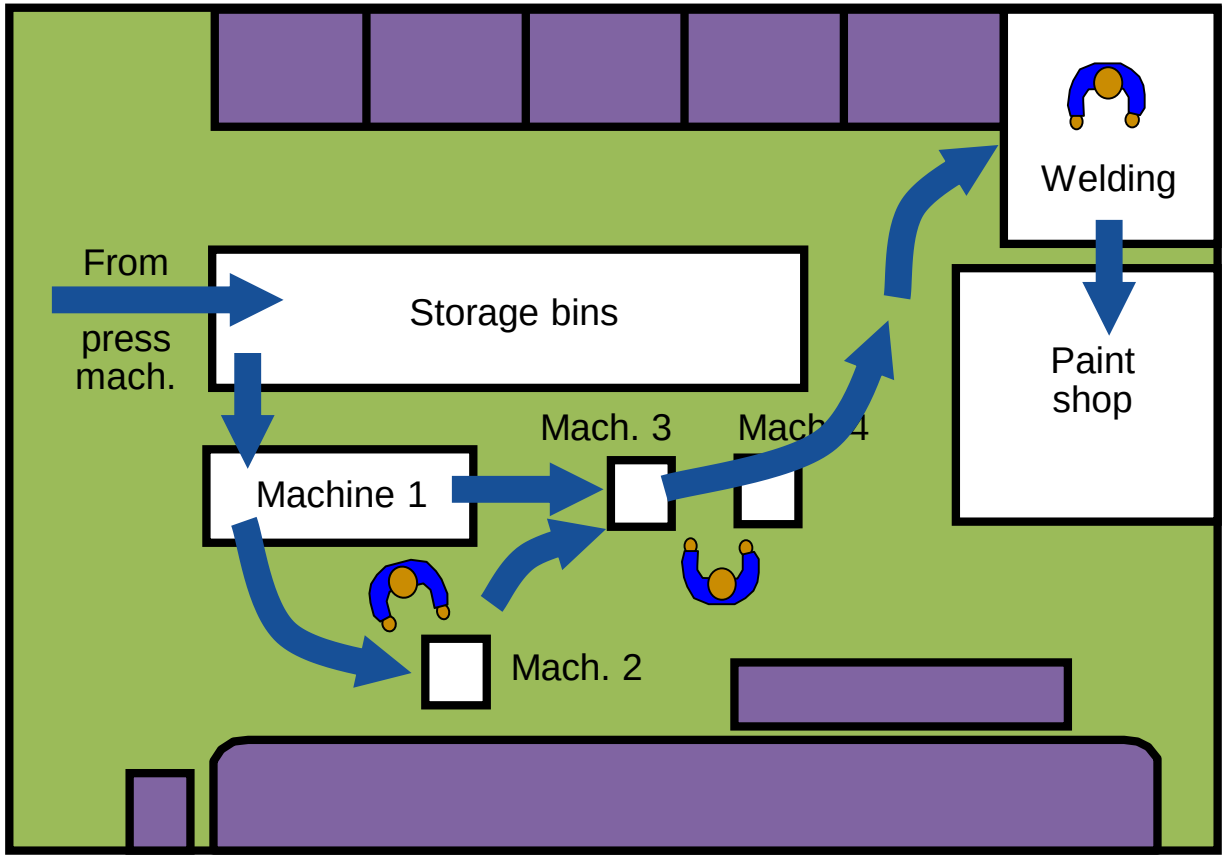

Figure 10.5 (a)

Flow Diagram

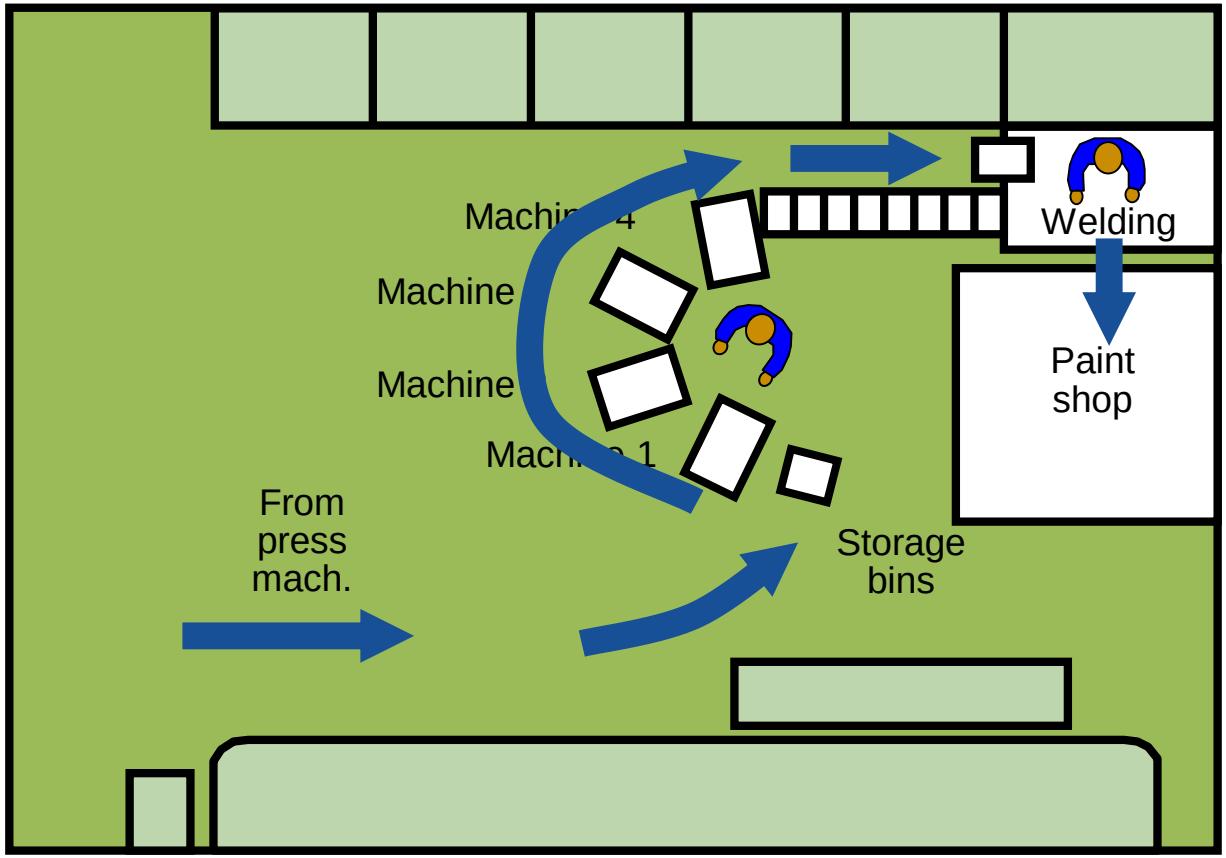

Figure 10.5 (b)

Process Chart

Present Method ☐
Proposed Method ☒

PROCESS CHART

SUBJECT CHARTED *Axle-stand Production*

DATE *5 / 1 / 15*
CHART BY *JH*
CHART NO. *1*
SHEET NO. *1* OF *1*

DEPARTMENT *Work cell for axle stand*

DIST. IN FEET	TIME IN MINS.	CHART SYMBOLS	PROCESS DESCRIPTION
50			*From press machine to storage bins at work cell*
	3		*Storage bins*
5			*Move to machine 1*
	4		*Operation at machine 1*
4			*Move to machine 2*
	2.5		*Operation at machine 2*
4			*Move to machine 3*
	3.5		*Operation at machine 3*
4			*Move to machine 4*
	4		*Operation at machine 4*
20			*Move to welding*
	Poka-yoke		*Poka-yoke inspection at welding*
	4		*Weld*
10			*Move to painting*
	4		*Paint*
97	25		TOTAL

○ = operation; ⇨ = transport; □ = inspect; D = delay; ▽ = storage

Activity Chart

	OPERATOR #1		OPERATOR #2	
	TIME	%	TIME	%
WORK	12	100	12	100
IDLE	0	0	0	0

OPERATION: Oil change & fluid check
EQUIPMENT: One bay/pit
OPERATOR: Two-person crew
STUDY NO.: _____ ANALYST: BR

SUBJECT *Quick Car Lube* DATE *5-1-15*
PRESENT / PROPOSED DEPT. SHEET *1* OF *1* CHART BY *LSA*

	TIME	Operator #1	TIME	Operator #2	TIME
	2	Take order		Move car to pit	
	4	Vacuum car		Drain oil	
	6	Clean windows		Check transmission	
	8	Check under hood		Change oil filter	
	10	Fill with oil		Replace oil plug	
	12	Complete bill		Move car to front for customer	
	14	Greet next customer		Move next car to pit	
Repeat cycle	16	Vacuum car		Drain oil	
	18	Clean windows		Check transmission	

Operations Chart

SYMBOLS	PRESENT		PROPOSED	
	LH	RH	LH	RH
○ OPERATION	2	3		
⇨ TRANSPORT.	1	1		
□ INSPECTION				
D DELAY	4	3		
▽ STORAGE				

PROCESS: Scooping Ice for Coffee
EQUIPMENT: Scoop
OPERATOR: Starbucks
STUDY NO: _____ ANALYST: CM
DATE: 5 /1 /15 SHEET NO. 1 of 2
METHOD (PRESENT / PROPOSED)
REMARKS: Partial Study

LEFT-HAND ACTIVITY Present METHOD	DIST.	SYMBOLS	SYMBOLS	DIST.	RIGHT-HAND ACTIVITY Present METHOD
1 Reach for cup		●⇨□D▽	○⇨□D▽		Idle
2 Grasp cup		●⇨□D▽	○⇨□D▽		Idle
3 Move cup	6"	○➡□D▽	○⇨□D▽		Idle
4 Hold cup		○⇨□D▽	●⇨□D▽		Reach for scoop
5 Hold cup		○⇨□D▽	●⇨□D▽		Grasp scoop
6 Hold cup		○⇨□D▽	○➡□D▽	8"	Move scoop to ice
7 Hold cup		○⇨□D▽	●⇨□D▽		Scoop ice

The Visual Workplace

- Use low-cost visual devices to share information quickly and accurately
- Displays and graphs replace printouts and paperwork
- Able to provide timely information in a dynamic environment
- System should focus on improvement

Examples of Visual Workplace

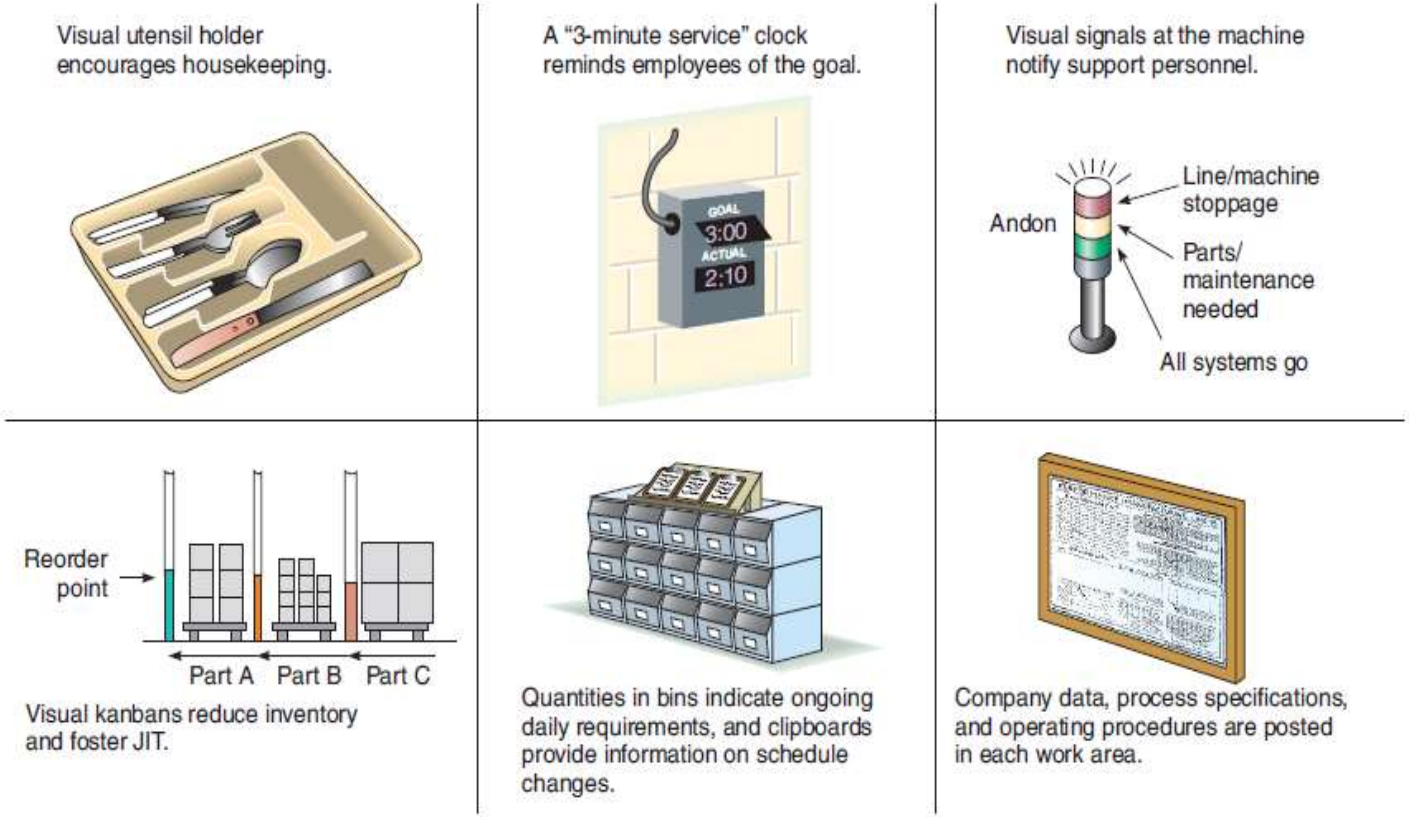

Figure 10.8

The Visual Workplace

Take Home Exercise

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- Read in detail from Your Text Book Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 10

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Any Question

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Labour Standards

Historical experiences, Time Studies, Predetermined standards, Work Sampling

Meaningful Standards Help Determine

1. Labor content of items produced
2. Staffing needs
3. Cost and time estimates
4. Crew size and work balance
5. Expected production
6. Basis of wage-incentive plans
7. Efficiency of employees

Labor Standards

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- The amount of time required to perform a job or part of a job.
- Established through :

1. Historical experience
2. Time studies
3. Predetermined time standards
4. Work sampling

Historical Experience

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- Estimated based on, how many labor hours were required to do a task the last time.
- It is relatively easy and inexpensive.
- Accuracy is questionable.

Time Studies

- Involves timing a sample of a worker's performance and using it to set a standard
- Requires trained and experienced observers
- Cannot be set before the work is performed

Time Studies

1. Define the task to be studied
2. Divide the task into precise elements
3. Decide how many times to measure the task
4. Time and record element times and rating of performance

Time Studies

5. Compute **average observed time**

$$\text{Average observed time} = \frac{\text{Sum of the times recorded to perform each element}}{\text{Number of observations}}$$

6. Determine performance rating and **normal time**

$$\text{Normal time} = \text{Average observed time} \times \text{Performance rating factor}$$

Time Studies

7. Add the normal times for each element to develop the total normal time for the task
8. Compute the **standard time**

$$\text{Standard time} = \frac{\text{Total normal time}}{1 - \text{Allowance factor}}$$

Class Exercise

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- The time study of a work operation at a Red Lobster restaurant yielded an average observed time of 4.0 minutes.
- The analyst rated the observed worker at 85%. This means the worker performed at 85% of normal when the study was made.
- The firm uses a 13% allowance factor.
- Red Lobster wants to compute the normal time and the standard time for this operation.

Solution

Stop Watch Time Study

1. Define the task to be studied, and inform the worker who will be studied.
2. Determine the number of cycles to observe.
3. Time the job (delete abnormal data if required) , and rate the worker's performance.
4. Compute the standard time.

Class Exercise

- Management Science Associates promotes its management development seminars by mailing thousands of individually composed and typed letters to various firms.
- A time study has been conducted on the task of preparing letters for mailing.
- On the basis of the following observations, Management Science Associates wants to develop a time standard for this task.
- The firm's personal, delay, and fatigue allowance factor is 15%.

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JOB ELEMENT	OBSERVATIONS (MINUTES)					PERFORMANCE RATING
	1	2	3	4	5	
(A) Compose and type letter	8	10	9	21*	11	120%
(B) Type envelope address	2	3	2	1	3	105%
(C) Stuff, stamp, seal, and sort envelopes	2	1	5*	2	1	110%

How many Readings You Required ??

Determination of Sample Size

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$$\text{Required sample size} = n = \left(\frac{zs}{h\bar{x}}\right)^2 \qquad (10\text{-}4)$$

h = accuracy level (acceptable error) desired in percent of the job element, expressed as a decimal (5% = .05)

z = number of standard deviations required for desired level of confidence (90% confidence = 1.65; see Table 10.2 or Appendix I for more z-values)

s = standard deviation of the initial sample

$\bar{x}$ = mean of the initial sample

n = required sample size

Class Exercise

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- Thomas W. Jones Manufacturing Co. has asked you to check a labor standard prepared by a recently terminated analyst. Your first task is to determine the correct sample size.
- Your accuracy is to be within {5% and your confidence level at 95%.
- The standard deviation of the sample is 1.0 and the mean 3.00.

Solution

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$$h = 0.05 \quad \bar{x} = 3.00 \quad s = 1.0$$

$$z = 1.96 \text{ (from Table 10.2 or Appendix I)}$$

$$n = \left(\frac{zs}{h\bar{x}}\right)^2$$

$$n = \left(\frac{1.96 \times 1.0}{0.05 \times 3}\right)^2 = 170.74 \approx 171$$

Predetermined Time Standards

- Published data based on extensive research to determine standard elemental times.
- Among the advantages of predetermined time standards are the following:

1. They are based on large numbers of workers under controlled conditions.
2. The analyst is not required to rate performance in developing the standard.
3. There is no disruption of the operation.
4. Standards can be established even before a job is done.

Work Sampling

The study material is prepared by Prof. Abhishek Maiti for the students of Operation Management

- An estimate, via sampling, of the percentage of the time that a worker spends on various tasks.

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$$n = \frac{z^2 p(1-p)}{h^2} \qquad (10\text{-}7)$$

n = required sample size

z = number of standard deviations for the desired confidence level ($z = 1$ for 68.27% confidence, $z = 2$ for 95.45% confidence, and $z = 3$ for 99.73% confidence—these values are obtained from Table 10.2 or the normal table in Appendix I)

p = estimated value of sample proportion (of time worker is observed busy or idle)

h = acceptable error level, in percent (as a decimal)

Class Exercise (Part a)

The study material is prepared by Prof. Abhishek Maiti for the students of Operation Management

- The manager of Michigan County's welfare office, Dana Johnson, estimates that her employees are idle 25% of the time.
- She would like to take a work sample that is accurate within {3% and wants to have 95.45% confidence in the results.

Solution

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n = required sample size

z = confidence level (2 for 95.45% confidence)

p = estimate of idle proportion = 25% = .25

h = acceptable error of 3% = .03

$$n = \frac{(2)^2(.25)(.75)}{(.03)^2} = 833 \text{ observations}$$

Class Exercise (Part b)

- Dana Johnson, the manager of Michigan County's welfare office, wants to be sure her employees have adequate time to provide prompt, helpful service.
- She believes that service to welfare clients who phone or walk in without an appointment deteriorates rapidly when employees are busy more than 75% of the time.
- Consequently, she does not want her employees to be occupied with client service activities more than 75% of the time.

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NO. OF OBSERVATIONS	ACTIVITY
485	On the phone or meeting with a welfare client
126	Idle
62	Personal time
23	Discussions with supervisor
137	Filing, meeting, and computer data entry
833	

Summary

Human Resource Strategy

- **Labour Planning** — Employment Stability, Work Schedules, Work Rules
- **Job Design** — Job Specialization, Job Expansion, Psychological components, Self Directed Teams, Motivation and Incentive System
- **Labour Standards** — Historical experiences, Time Studies, Predetermined standards, Work Sampling

Further Reading / References

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1. Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 10
2. Operation Management by William J Stevenson—Chapter 7
3. HS Talks –Job Design.

THANK YOU